

CM0845 Logic

First-Order Logic: Syntax

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Remark

The references for this section are van Dalen (2013, § 3.1, § 3.2 and § 3.3).

Introduction

Example

Informal examples (van Dalen 2013, p. 53):

$\exists x P(x)$ there is an x with property P

$\forall y P(y)$ for all y P holds (all y have the property P)

$\forall x \exists y (x = 2y)$ for all x there is a y such that x is two times y

$\forall \epsilon (\epsilon > 0 \rightarrow \exists n (n < \epsilon))$ for all positive there is an n such that $n < \epsilon$

$x < y \rightarrow \exists z (x < z \wedge z < y)$ if $x < y$, then there is a z such that $x < z$ and $z < y$

$\forall x \exists y (x \cdot y = 1)$ for each x there exists an inverse y

Structures

Definition

A **structure** is an ordered sequence

$$\langle A, R_1, \dots, R_n, F_1, \dots, F_m, \{c_i \mid i \in I\} \rangle,$$

where

- (i) A is a non-empty set, the **universe** of the structure,
- (ii) $R_1, \dots, R_n$ are relations on A ,
- (iii) $F_1, \dots, F_m$ are functions on A , and
- (iv) the c_i , where $i \in I$, are elements of A (constants).

Notation

- Structures are denoted by Gothic capitals: $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \dots$
- $A = |\mathfrak{A}|$

Structures

Examples

Whiteboard.

Definition

The **similarity type** (or **signature** or **non-logical constants**) of a structure

$$\langle A, R_1, \dots, R_n, F_1, \dots, F_m, \{c_i \mid i \in I\} \rangle$$

is a sequence

$$\langle r_1, \dots, r_n; a_1, \dots, a_m; \kappa \rangle,$$

where

- (i) $R_i \subseteq A^{r_i}$,
- (ii) $F_j : A^{a_j} \rightarrow A$, and
- (iii) $\kappa = |\{c_i \mid i \in I\}|$ (cardinality of I).

Structures

Examples

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Structures

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Limiting cases

0-ary relations and 0-ary functions.

Structures

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Limiting cases

0-ary relations and 0-ary functions.

Convention

All the structures are equipped implicitly with the identity relation.

Alphabet

Definition

The **alphabet** has the following symbols:

- (i) Predicate symbols: $P_1, \dots, P_n$ and $\doteq$
- (ii) Function symbols: $f_1, \dots, f_m$
- (iii) Constant symbols: $\bar{c}_i$ for $i \in I$
- (iv) Variables: $x_0, x_1, x_2, \dots$ (countably many)
- (v) Connectives: $\vee, \wedge, \rightarrow, \neg, \leftrightarrow, \perp, \forall, \exists$
- (vi) Auxiliary symbols: $(,)$

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Remark

The equality symbol.

The Set of Terms

Definition

The **set of terms**, denoted **TERM**, is the **smallest** set X with the properties:

- (i) $x_i \in X$, where $i \in \mathbb{N}$,
- (ii) $\bar{c}_i \in X$, where $i \in I$, and
- (iii) $t_1, \dots, t_{a_i} \in X \Rightarrow f_i(t_1, \dots, t_{a_i}) \in X$, for $1 \leq i \leq m$.

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Examples

Whiteboard.

The Set of Formulae

Definition

The **set of formulae**, denoted **FORM**, is the **smallest** set X with the properties:

- (i) $\perp \in X$,
- (ii) $P_i \in X$ if $r_i = 0$,
- (iii) $t_1, \dots, t_{r_i} \in \text{TERM} \Rightarrow P_i(t_1, \dots, t_{r_i}) \in X$,
- (iv) $t_1, t_2 \in \text{TERM} \Rightarrow t_1 \doteq t_2 \in X$,
- (v) $\varphi, \psi \in X \Rightarrow (\varphi \square \psi) \in X$, where $\square \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$,
- (vi) $\varphi \in X \Rightarrow (\neg \varphi) \in X$,
- (vii) $\varphi \in X \Rightarrow ((\forall x_i)\varphi) \in X$ and
- (viii) $\varphi \in X \Rightarrow ((\exists x_i)\varphi) \in X$.

The formulae defined in the four first items are called **atomic formulae** or **atoms**.

Notational Conventions

- We use the conventions of propositional logic.
- We delete the outer brackets and the brackets round $\forall x$ and $\exists x$ whenever possible.
- Quantifiers bind more strongly than binary connectives.
- Join strings of quantifiers, e.g. $\forall x_1 x_2 \exists x_3 x_4 \varphi$ stands for $\forall x_1 \forall x_2 \exists x_3 \exists x_4 \varphi$.

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Examples

Whiteboard.

Induction Principles and Recursive Definitions

Remark

Given that **TERM** and **FORM** are set inductively defined, we have induction principles and recursive definitions on them.

Set of Free Variables of a Term

Definition

The **set of free variables of a term** t , denoted $FV(t)$, is defined by

$$\begin{aligned} FV : \text{TERM} &\rightarrow \{x_i \mid i \in \mathbb{N}\} \\ FV(x_i) &= \{x_i\}, \\ FV(\bar{c}_i) &= \emptyset, \\ FV(f(t_1, \dots, t_n)) &= FV(t_1) \cup \dots \cup FV(t_n). \end{aligned}$$

Closed Terms

Definition

A term t is **closed** iff $FV(t) = \emptyset$. The set of closed terms is denoted by $TERM_c$.

Examples

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Set of Free Variables of a Formula

Definition

The **set of free variables of a formula** φ , denoted $FV(\varphi)$, is defined by

$$FV : \text{FORM} \rightarrow \{x_i \mid i \in \mathbb{N}\}$$

$$FV(\perp) = \emptyset$$

$$FV(P) = \emptyset, \text{ for } P \text{ propositional symbol}$$

$$FV(P(t_1, \dots, t_n)) = FV(t_1) \cup \dots \cup FV(t_n),$$

$$FV(t_1 \doteq t_2) = FV(t_1) \cup FV(t_2),$$

$$FV(\varphi \square \psi) = FV(\varphi) \cup FV(\psi),$$

$$FV(\neg\varphi) = FV(\varphi),$$

$$FV(\forall x_i \varphi) = FV(\exists x_i \varphi) = FV(\varphi) - \{x_i\}.$$

Sentences

Definition

A formula φ is **closed** iff $FV(\varphi) = \emptyset$. A closed formula is also called a sentence. The set of sentences is denoted by **SENT**.

Examples

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Free Terms for a Variable in a Formula

Definition

A **term** t is **free for a variable** x in a formula φ iff

- (i) φ is atomic,
- (ii) $\varphi := \neg\psi$ and t is free for x in ψ ,
- (iii) $\varphi := \varphi_1 \Box \varphi_2$ and t is free for x in φ_1 and φ_2 ,
- (iv) $\varphi := \forall y\psi$ and if $x \in \text{FV}(\varphi)$, then $y \notin \text{FV}(t)$ and t is free for x in ψ , or
- (v) $\varphi := \exists y\psi$ and if $x \in \text{FV}(\varphi)$, then $y \notin \text{FV}(t)$ and t is free for x in ψ .

The Extended Language

Definition

The **extended language**, $L(\mathfrak{A})$, of $\mathfrak{A}$ is obtained from the language L , of the type of $\mathfrak{A}$, by adding constant symbols for all elements of $|\mathfrak{A}|$. We denote the constant symbol, belonging to $a \in |\mathfrak{A}|$, by $\bar{a}$.

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Examples

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References

-  Dirk van Dalen [1980] (2013). Logic and Structure. 5th ed. Springer (cit. on pp. 2, 3).